

BEE6940 Final Project: Forest Fire Modeling

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Modeling Process

Data Description/Visualization

As an initial pass, all forest fires in our dataset were spatially plotted to understand general patterns and distribution (**Figure 1**); K-Means clustering was then used to partition the data points into groups (**Figure 2**). Based on the inertia of each k-means run over various cluster sizes ($k=1$ to 20), having seven clusters was determined to be an appropriate cluster size, with the results shown in Figure 1. The clusters are roughly partitioned into regions across the United States (for instance, most forest fires identified in Cluster 7 are located within the south-east part of the country). Figure 2b shows the proportion of each cluster that corresponds to big fires (fires that spread larger than 100 acres): both clusters 3 and 6 have the highest proportions at 0.48 and 0.41 respectively and are generally located within the western half of the United States.

A heatmap visualization of the variance covariance matrix revealed similar findings about the relationships between location and the meteorologic variables and the occurrence of 100+ acre wildfires (**Figure 3**). Notably, the 3 leading measures of humidity and longitude appeared to have the highest correlation `big_fire` predictand variable. This is not a particularly surprising result given the importance of fuel moistures and fuel type on rates of fire spread. From our initial analysis, it seems that the spatial location of a forest fire has a noticeable impact on the likelihood of big fires, though these spatial tendencies could be driven by the meteorological variables as well; further reasoning is provided in our Explainability section.

The forest fire data indicate an imbalanced classification problem: only 18% of the data is labeled as a big fire, which adds some challenge in training our models. Accuracy is less useful as a metric in this case, and so we use other metrics such as the recall, F1 and AUC scores, which can provide a better metric for model performance in this context.

Models

As a baseline, a simple logistic regression was fitted and tuned, yielding an F1 score of 0.88 for small fires, 0.31 for big fires, and an AUC score of .82. Working from this baseline, three different machine learning architectures were implemented: the MLP classifier neural network, the Random Forest classifier, and a boosted trees classifier (XGBoost). For all architectures, the forest fire dataset was split into training, validation, and test sets (80%, 10%, 10% of the original dataset respectively).

For the MLP classifier, a GridSearch with cross-validation (five total folds) was performed in order to identify the best parameters for our data. Of particular interest here was the total number of hidden layers: up to three total hidden layers (with different size ratios amongst the layers) were tested within our Gridsearch framework, along with two activation functions (ReLU and tanh), multiple regulation parameters, and both adaptive and constant learning rates. An

additional GridSearch was run after the first to further fine-tune the hidden layer sizes. As part of the tuning process, a synthetic minority over-sampling technique (SMOTE) was applied to the dataset in an attempt to manage the classification imbalance, which allows for over-sampling of our minority class (in this case, big fires). Application of SMOTE did lead to improved AUC and F1 scores within our test set. Our Gridsearch indicated that two hidden layers of the same size (50 each) with tanh as an activation function, adaptive learning rate, and a regulation parameter of 0.05 provided the best results.

For the Random Forest classifier, a preliminary random forest model was constructed with square root chosen for the max features to help with generalization, a balanced class weight since the frequency in the dataset was notably unbalanced, and baseline values for the other hyperparameters. Stratified k-fold cross validation was also utilized to maintain the distribution of big fires in the folds so that the disproportionality is accounted for, ensuring more representative performance estimates. Following this preliminary model, multiple trends were investigated in an effort to identify potential relationships between variables. Among these were humidity and temperature trends which were then added to the model. Running the same random forest on this augmented data yielded improved AUC and F1 scores. A GridSearch with cross-validation was then performed to optimize the preliminarily estimated hyperparameters. Following the GridSearch, the results indicated that unrestricted max depth, a minimum of 5 samples in a leaf, and 350 estimators produced the best cross-validated AUC. Both the original model and the tuned model were evaluated using the previously described train, validation, and test split to compare model performances. Multiple GridSearches were performed to test a wide range of hyperparameters to ensure that parameter selection was not overly sensitive.

The boosted tree model XGBoost, was also implemented with similar grid-search protocols, optimized for f1 score as well as a positive scaling argument to better handle the imbalance of the dataset. The XGBoost model grid-search ultimately selected a learning rate of 0.1, max depth of 7, number of estimators (sequential trees) of 100, and a subsample size of 0.7.

Model Results

Our model performances for big forest fire prediction based on the test dataset are provided below (**Figure 4**). All three of our advanced models overall performed better than the baseline logistic regression model, though to varying extents. In general, the F1 scores were hard to improve on, being a combination of both the precision and recall; because of the classification imbalance, these values often suffer when attempting to predict big fires and can lead to a low F1 score. Nevertheless, our fine-tuning yielded better prediction ability, in particular with the feature engineering Random Forest and XGBoost models.

From these results, we decided on selecting our XGBoost model as our best model and the one we are proceeding with. When comparing the AUCs and F1 scores of the feature-engineered Random Forest and XGBoost models to one another, there was minimal difference between the two. The tie-breaking factor in selecting between the two was the recall results for predicting big forest fires (**Figure 5**). For predicting big forest fires, the

feature-engineered Random Forest model had a recall of 0.59 while the XGBoost model had a recall of 0.73. In this specific instance, we propose that there should be a priority on accurately predicting big wildfires when they are present. The feature-engineered Random Forest model performed better in balancing between precision and recall, but the XGBoost performed with a notably greater recall and that was ultimately what led to its selection as our final model.

Explainability

Multiple visualization methods were implemented to identify the drivers of large wildfire occurrences. A feature importance plot was developed based on the baseline random forest model (**Figure 6**). Of note is that longitude and latitude have the highest variable importance, followed by the three humidity-based variables. This was further substantiated as we removed latitude and longitude to explore their overall impact on the Random Forest model; doing so resulted in significantly lower classification metrics. This indicates that spatial variables capture important information that is not fully explained by the metrological variables alone. These unobserved variables could include differences in vegetation, topography, current and historic land management practices, access and infrastructure, availability of suppression personnel, and likely more beyond. Based on our exploratory analysis into feature explainability, the results indicate that the order of general feature importance goes spatial position, followed by humidity, then temperature, wind, and finally precipitation.

Among the meteorological inputs, the range of humidity variables are the most important to our models, which is not surprising due to the aforementioned importance of fuel moistures and dry vegetation to flammability and wildfire spread. Temperature contributes by increasing evaporation rate which leads to drier vegetation. Wind impacts fire spread and fire growth due to the increased oxygen supply and distribution of embers. An interesting finding in the set of wind variables was that wind 30 had a higher impact than wind 15 or 7. We suspect this could be due to the evaporative power of wind contributing to fuel treatment and drying. Finally, precipitation influences fuel dryness as elevated precipitation rehydrates fuel and thus limits fire spread, and lower precipitation, more specifically, prolonged low precipitation, leads to dry periods which increases the risk of big wildfires. It is interesting that precipitation is not a more powerful predictor, but after reviewing the SHAP values from the XGBoost model (**Figure 7**) we suggest that limited variation of the precipitation variable across the entire dataset may account for its lack of influence in the model.

These drivers all make physical sense as they are fundamentally important to how large wildfires can start and grow, with fuel moisture, atmospheric conditions, and fire spread contributing to determine the size and severity of wildfires.

The last variable that warrants discussion is the distance from a meteorological site. The variable shows significant model influence, below only the humidity and location variables. We suggest this could also be capturing variation from unobserved variables, as well as uncertainty around the inputs from localized weather events like wind and precipitation. For example, if an observation is far from the site where meteorological features are observed, it may be more likely to lean on an input like geographic location or humidity which are less dependent on exact location and are highly generalizable.

Figures

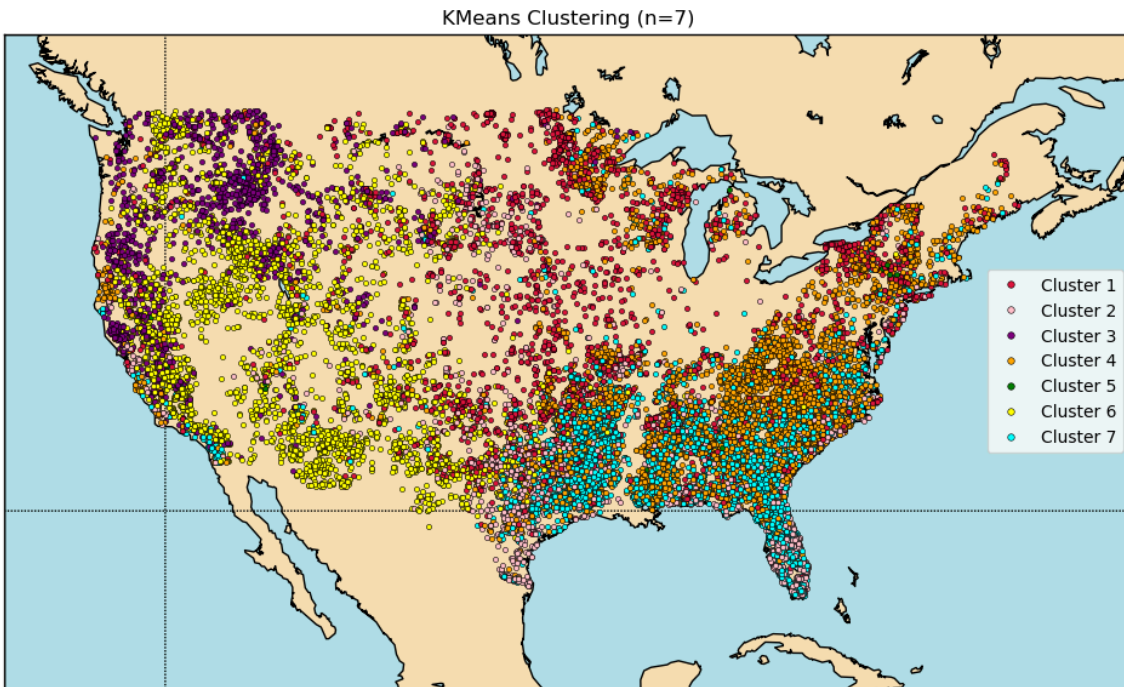


Figure 1. Spatial distribution of recorded forest fires in the United States ($n=24,637$) with k -means clustering applied ($k=7$).

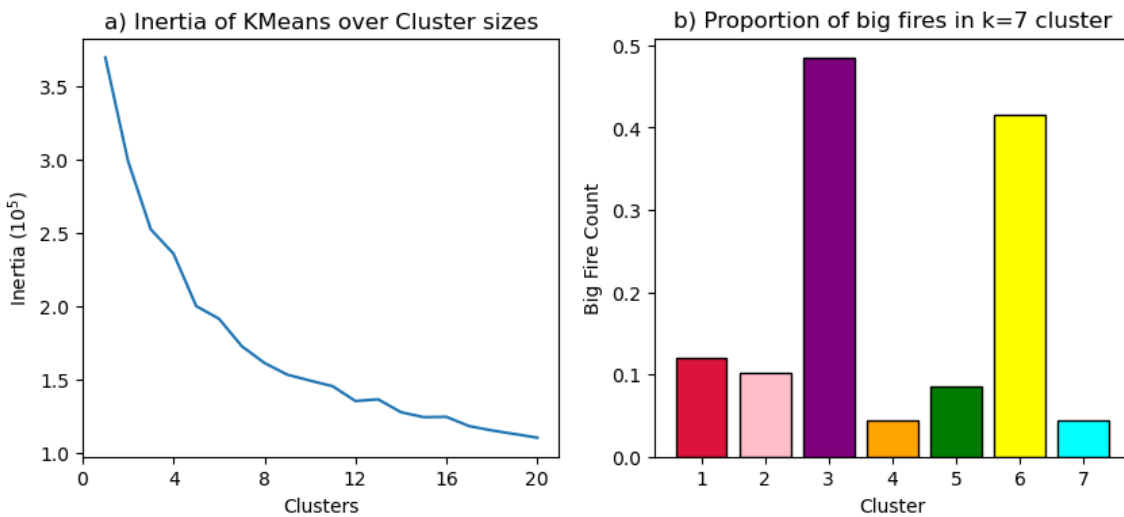


Figure 2. a) The inertia of k -means over cluster sizes from 1 to 20; b) The proportion of data points that correspond to a big fire in each of the seven clusters from our final k -means analysis.

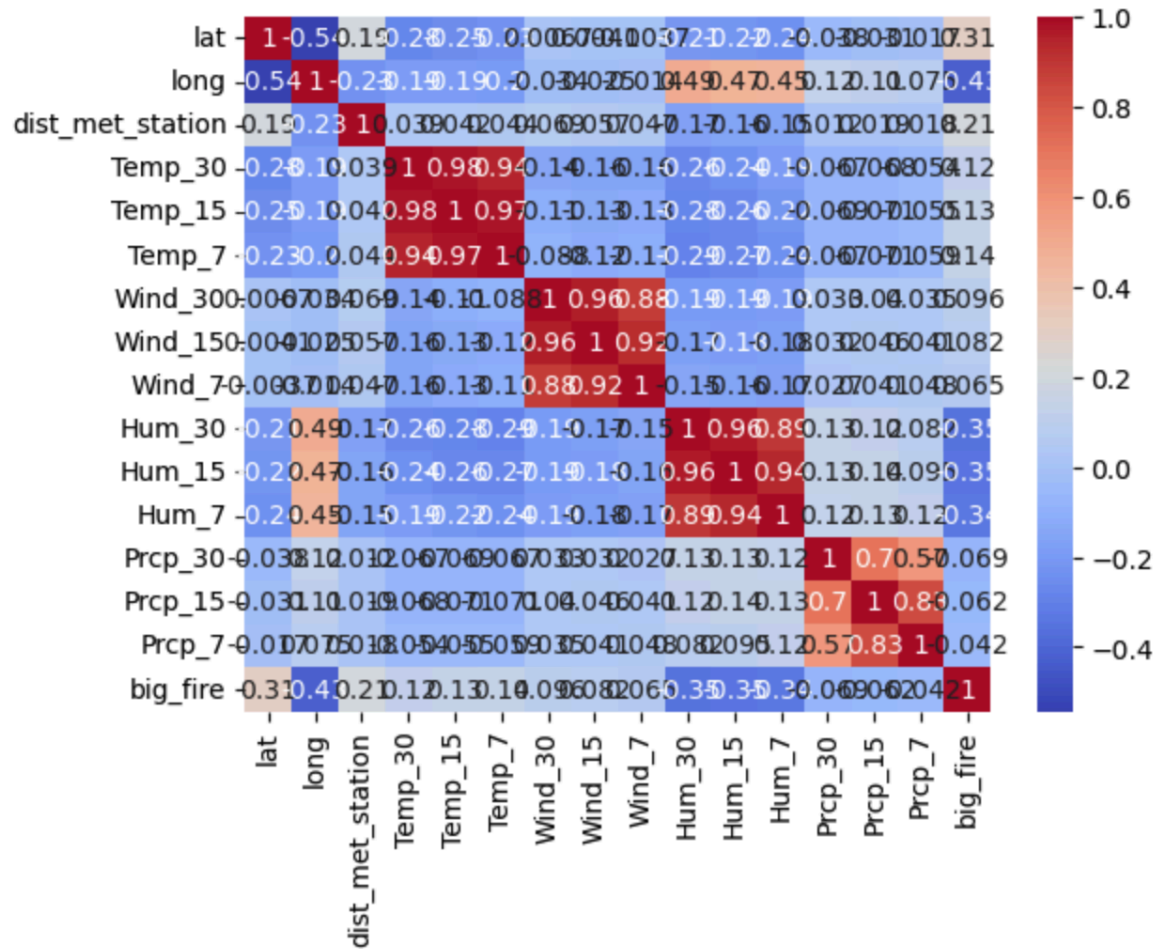


Figure 3. Heatmap of our calculated variance-covariance matrix for the entire forest fire dataset.

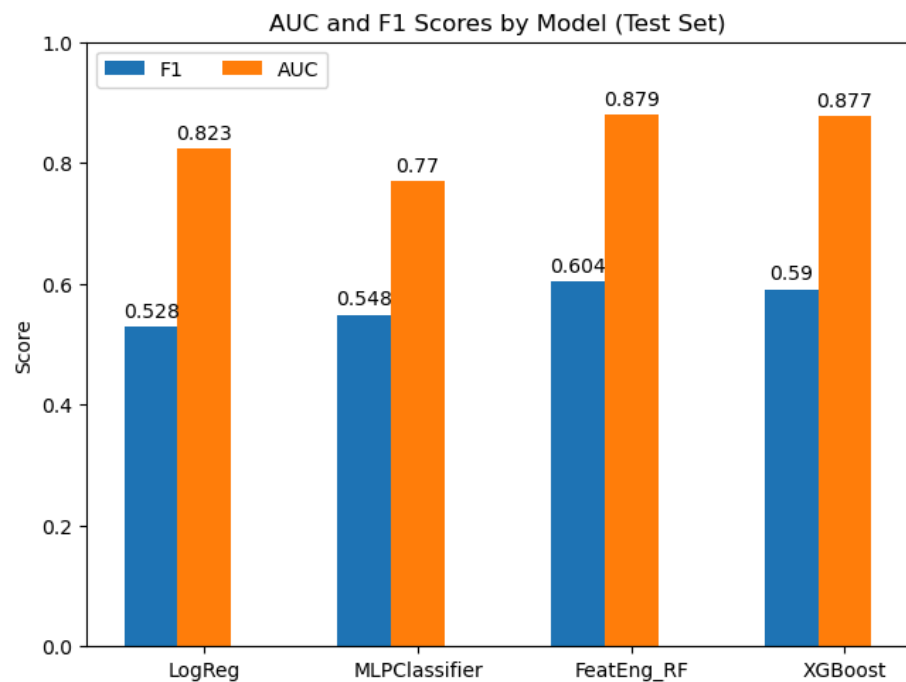


Figure 4. AUC (in orange) and F1 (in blue) scores using the test dataset across our four trained models.

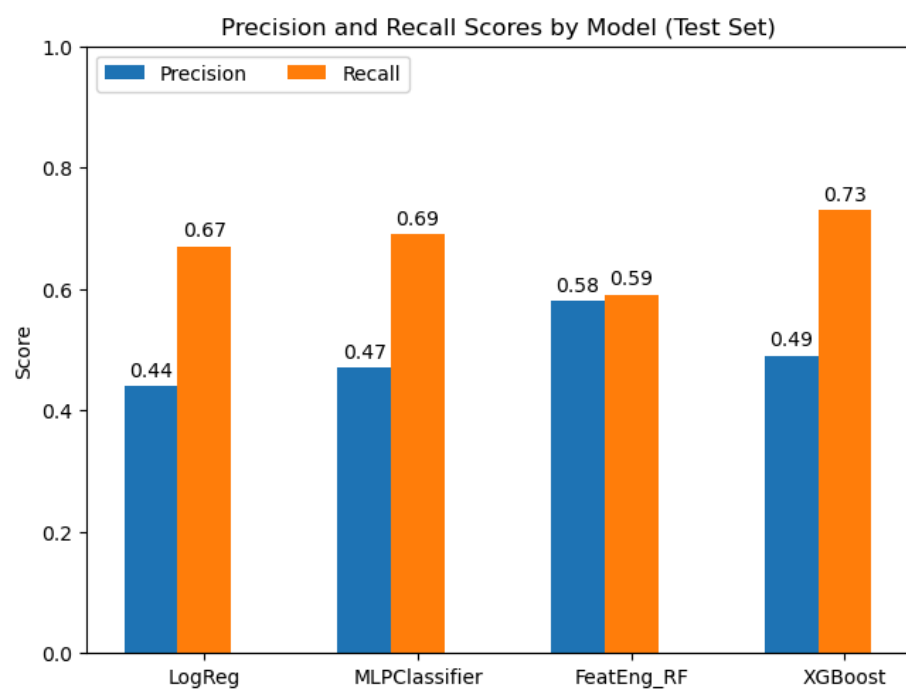


Figure 5. Precision (in blue) and recall (in orange) scores using the test dataset across our four trained models.

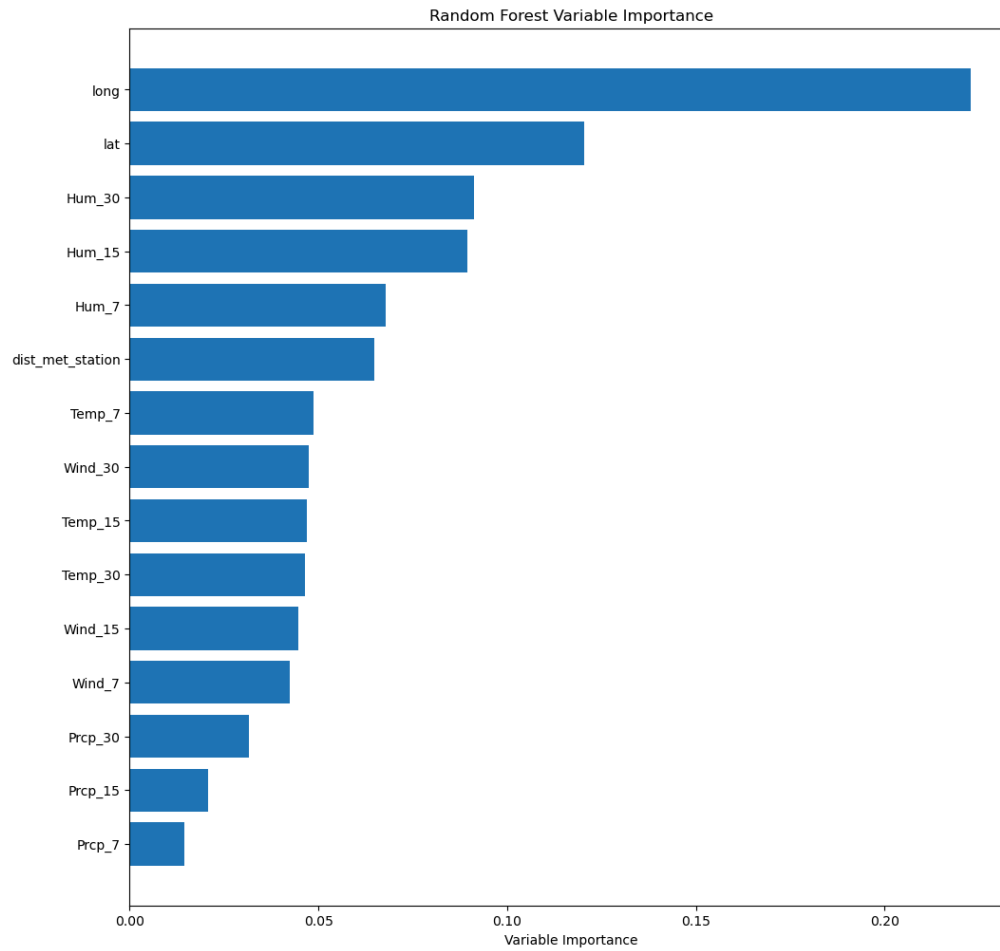


Figure 6. *Rankings of feature importance from the baseline Random Forest Model.*

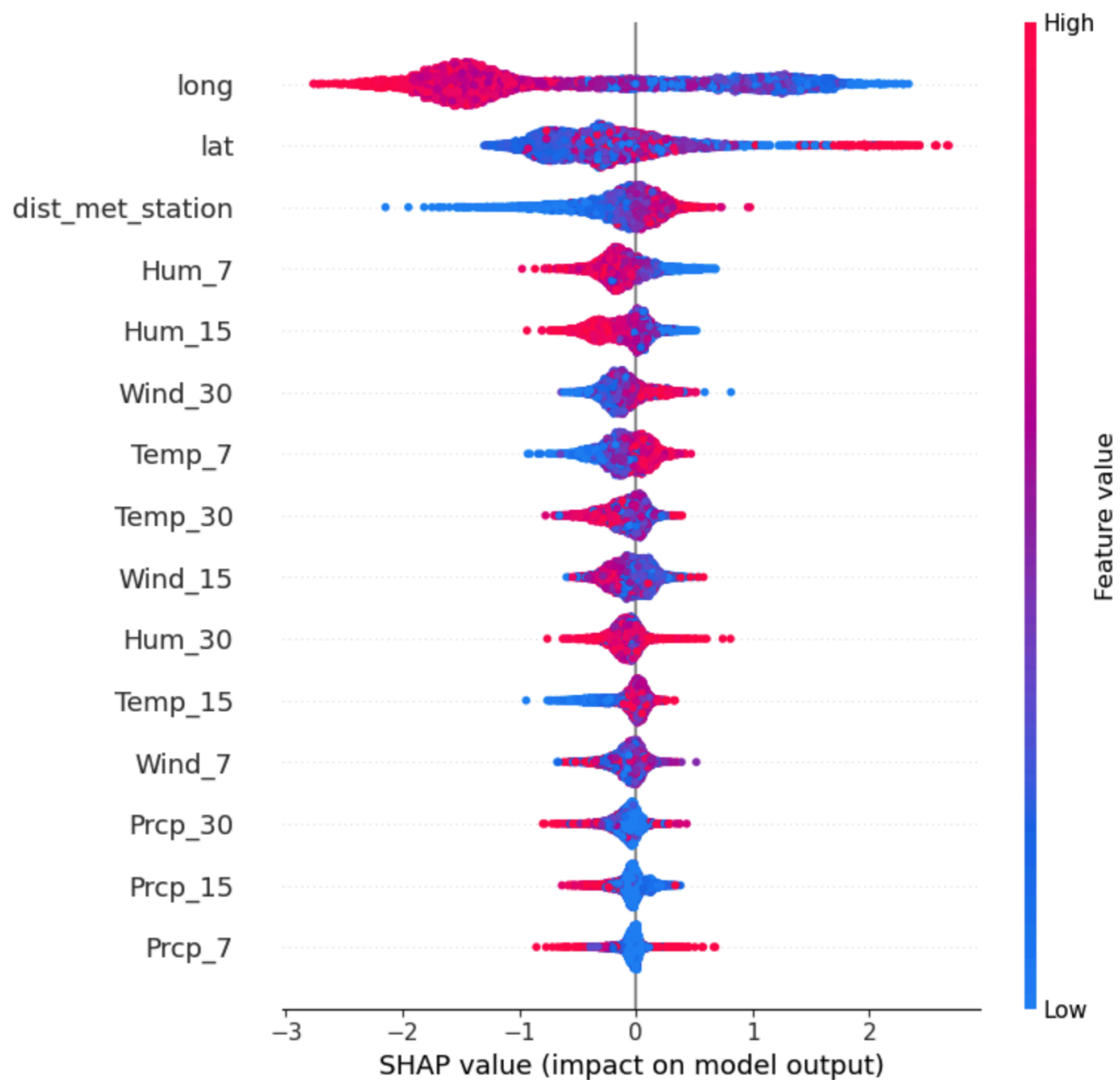


Figure 7. The SHAP values from the XGBoost model.